

KEETMANSHOOP, Namibia

WMO: 683120

Lat: 26.540S

Lon: 18.111E

Elev: 3506

StdP: 12.93

Time Zone: 2.00 (E02)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.6	40.6	5.0	8.1	65.1	8.8	9.8	66.0	23.8	67.4	21.2	66.6	4.6	70	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.1	101.4	63.3	99.1	62.6	97.2	62.3	69.8	85.9	68.4	84.5	67.3	83.7	8.2	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	109.5	73.4	64.2	102.4	74.6	62.2	95.5	75.7	36.1	84.2	34.9	84.8	33.9	82.4	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.0	19.1	16.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.5	82.5	81.6	78.5	71.9	66.2	59.0	58.6	61.2	68.2	73.5	76.6	80.7
	DBStd	10.63	4.79	4.57	5.66	6.40	7.07	6.22	6.67	7.20	8.77	7.58	6.41	5.08
	HDD50	28	0	0	0	0	0	9	13	6	0	0	0	0
	HDD65	754	0	0	4	19	74	190	212	159	70	22	4	0
	CDD50	7872	1007	885	883	656	503	279	279	354	548	728	799	951
	CDD65	3124	542	465	422	225	112	10	13	43	167	286	353	486
	CDH74	38534	7222	5714	4803	2163	1084	183	226	578	1972	3577	4567	6445
CDH80	20140	4297	3129	2419	860	330	14	32	136	902	1820	2466	3735	
Wind	WSAvg	7.2	7.9	5.9	5.8	5.9	6.0	7.7	7.4	7.9	7.5	8.0	8.3	8.3
Precipitation	PrecAvg	5.9	1.1	1.5	1.3	0.6	0.2	0.1	0.0	0.0	0.1	0.2	0.3	0.4
	PrecMax	19.0	8.7	5.7	5.5	2.4	1.5	0.6	0.8	0.2	1.9	1.6	1.9	1.5
	PrecMin	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.6	1.6	1.5	1.2	0.6	0.4	0.1	0.2	0.0	0.4	0.4	0.4	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.2	102.1	99.4	95.3	90.9	82.1	83.8	87.3	97.1	100.3	103.0	103.5
		MCWB	64.2	62.4	62.3	61.1	58.2	53.5	52.9	55.1	58.8	61.2	62.6	63.6
	2%	DB	100.9	98.8	96.1	91.7	86.3	78.6	79.1	83.7	93.0	96.8	98.9	100.5
		MCWB	63.9	62.8	62.9	60.6	56.6	52.4	51.2	53.1	58.4	60.3	62.2	62.8
	5%	DB	98.8	96.5	93.5	88.3	83.3	76.0	75.8	80.4	89.3	93.5	96.3	98.2
		MCWB	63.0	62.8	62.7	60.0	55.9	51.9	50.6	51.8	56.8	59.6	61.9	62.4
	10%	DB	96.5	94.0	91.1	85.4	79.9	73.0	72.7	76.9	85.5	89.9	93.0	95.4
		MCWB	62.7	63.0	62.2	59.5	55.0	50.8	49.3	50.6	55.8	58.7	59.5	61.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	71.3	70.6	66.7	60.3	58.2	54.8	56.2	62.0	67.5	71.7	69.4
		MCDB	85.2	85.5	82.8	80.2	81.0	72.5	78.2	82.3	90.4	90.4	94.1	88.0
	2%	WB	68.4	69.6	68.6	64.9	58.4	54.7	52.7	54.2	59.6	64.5	67.7	67.5
		MCDB	84.6	84.9	82.0	76.9	80.6	72.8	75.3	79.1	87.1	85.9	87.6	86.4
	5%	WB	67.0	68.5	67.3	63.7	57.1	53.0	51.3	52.7	57.9	62.1	64.4	66.0
		MCDB	86.1	84.2	80.6	77.0	79.7	72.1	73.7	77.1	86.0	86.9	86.1	86.0
	10%	WB	65.7	67.3	66.2	62.4	55.8	51.5	49.9	51.2	56.2	60.2	62.0	64.6
		MCDB	87.4	83.3	79.8	77.9	77.3	70.7	71.1	74.7	83.0	84.9	86.9	88.1
Mean Daily Temperature Range	5% DB	MDBR	27.1	24.2	23.3	23.3	24.1	24.9	25.6	26.9	27.0	28.2	29.2	28.4
		MCDBR	28.9	27.1	25.6	26.7	26.3	27.0	28.6	29.4	29.7	31.5	31.9	30.1
		MCWBR	9.5	9.4	8.9	10.5	10.8	12.1	13.3	13.3	12.3	13.1	14.8	9.8
	5% WB	MCDBR	25.0	21.9	21.9	20.5	24.9	25.5	26.4	28.1	28.3	28.9	28.3	26.1
		MCWBR	8.7	8.3	7.2	7.7	10.5	12.5	12.5	12.8	12.6	13.9	15.7	10.1
Clear-Sky Solar Irradiance	taub	0.333	0.344	0.328	0.302	0.272	0.257	0.260	0.290	0.327	0.340	0.305	0.312	
	taud	2.520	2.500	2.521	2.571	2.623	2.626	2.605	2.470	2.345	2.343	2.528	2.540	
	Ebn at Noon	320	312	307	302	299	298	300	301	302	309	327	327	
	Edh at Noon	36	36	33	29	24	23	25	31	39	41	35	35	
All-Sky Solar Radiation	RadAvg	2619	2379	2094	1737	1450	1306	1388	1675	2037	2377	2672	2765	
	RadStd	111	143	76	63	39	35	30	54	65	74	78	88	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page